

Unit - 4(Plant Physiology) - 200 MCQs

NCERT Nichod : Biology Practice Series NEET 2025



Good evening
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Ready?

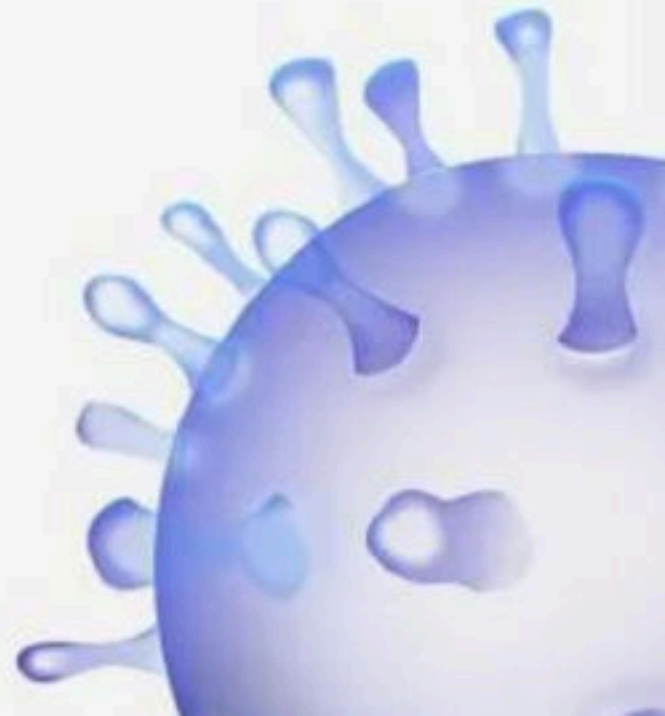


" 200 MCQS : Plants Physiology "

3 chapter :-
- Photosynth
- Resp
- Plant



BIOLOGYLIVE



① ^{V. Imp} During light reaction in photosynthesis the following are formed:-

- (1) ATP and sugar
- (2) Hydrogen, O_2 and sugar
- ✓ (3) ATP, hydrogen donor and O_2
- (4) ATP, hydrogen and O_2 donor



② What causes the decrease in rate of photosynthesis after exposing plant to higher light intensities?

- (1) Carotenoids block photosynthesis pathway
- ✓ (2) Destruction of chlorophyll molecule due to photooxidation
- (3) Change in leaf angle
- (4) Given question is wrong.



3. Identify the wrong statements among given statements.

(1) Water is one of the reactants in the light reaction

✓ (2) Water stress causes the stomata to close hence
reducing the CO_2 availability

(3) Water stress also makes leaves wilt

(4) The effect of water as a factor is more through its effect on the plant, rather than directly on photosynthesis.



④ Chemiosmosis requires ∴

- (1) Membrane ✓
- (2) Proton pump ✓
- (3) Proton gradient ✓
- ✓ (4) All of them



5 Statement A : $C_6H_{12}O_6$ represents glucose. ✓ ✓ H_2O
Statement B : The O_2 released is from carbon dioxide, this
was proved using radio isotope techniques ✗

- ✓ (1) Statement-A is correct but Statement-B is incorrect
(2) Statement-A is incorrect but Statement-B is correct
(3) Both the statements are correct
(4) Both the statements are incorrect



⑥ The electrons in the reaction centre of PS-I are

- (1) Excited simultaneously with PS-II
- (2) Excited simultaneously with P680
- (3) Excited simultaneously with P700 ✗
- (4) Both 1 and 2



7. Conversion of Phosphoenol pyruvic acid to pyruvic acid and ADP to ATP are examples of

- (1) Photophosphorylation.
- (2) Oxidative phosphorylation.
- (3) Photoelectric phosphorylation.
- ✓ (4) Substrate level phosphorylation.



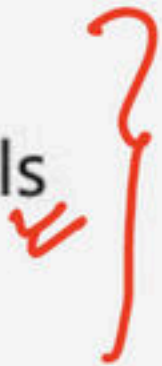
⑧ It is expected that CO₂ concentration could reach to a level of about 600 ml.L⁻¹ by 2020. In such case, the C₃ plant are likely to be benefitted more than the C₄ plant.

- ✓ (1) C₃ plants, C₄ plants
- (2) C₃ plants, C₂ plants
- (3) CAM plants, C₄ plants
- (4) C₄ plants, CAM plants



9. By looking at which internal structure of a plant can you tell whether a plant is C₃ or C₄?

- (1) Kranz anatomy ✓
- (2) Distribution of mesophyll cells ✓
- (3) Bundle sheath cells ✓
- (4) All the above ✓



10. Plants that are adapted to dry tropical regions have?

- (1) C_3 cycle
- ✓ (2) C_4 cycle
- (3) C_2 cycle
- (4) Both 2 and 3



11 Chemosynthetic bacteria obtain energy from :-

- (1) Sun
- (2) Infra-red rays
- (3) Organic substances
- ✓ (4) Inorganic chemicals



12. During photorespiration, the oxygen consuming reactions occurs in

- ✓ (1) Stroma of chloroplast and peroxisomes
- (2) Grana of chloroplast and peroxisomes
- (3) Stroma of chloroplast
- (4) Stroma of chloroplasts and mitochondria



13. Photophosphorylation in chloroplast is most similar to the

(1) Transcription in nucleus

✓(2) Mitochondrial oxidative phosphorylation

(3) Mitochondrial glycolysis

(4) Ribosomal translation



14 Statement A : Within the chloroplast, protons in the stroma decrease in number, while in the lumen there is accumulation of protons

Statement B : Accumulation of protons in the lumen creates a proton gradient across the thylakoid membrane as well as a measurable decrease in pH in the lumen

- (1) Statement-A is correct but Statement-B is incorrect
- (2) Statement-A is incorrect but Statement-B is correct
- ☒ (3) Both the statements are correct
- (4) Both the statements are incorrect



15. Which of the following is impact of water stress on plant?

- (1) Wilting of leaves ✓
- (2) Reducing of surface area of leaf ✓
- (3) Reducing the metabolic activity of plant ✓
- ✓ (4) All of the above



6. In Calvin cycle, RuBisCO incorporates CO_2 into ribulose, 1, 5-bisphosphate which rapidly splits into

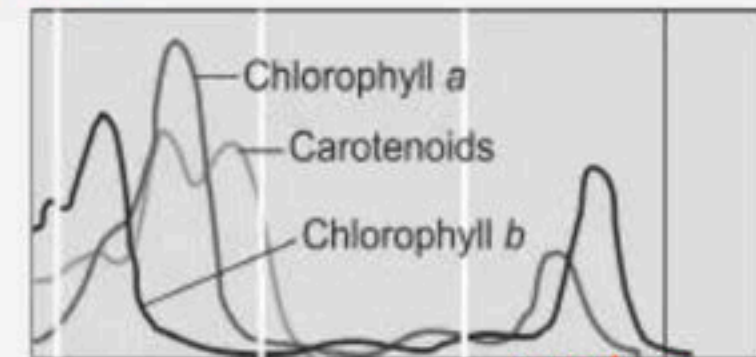
v. imp

- ✓ (1) 2 molecules of 3-PGA
- (2) 2 molecules of 2 -PGA
- (3) 3 molecules of 3 -PGA
- (4) 3 molecules of 2-PGA

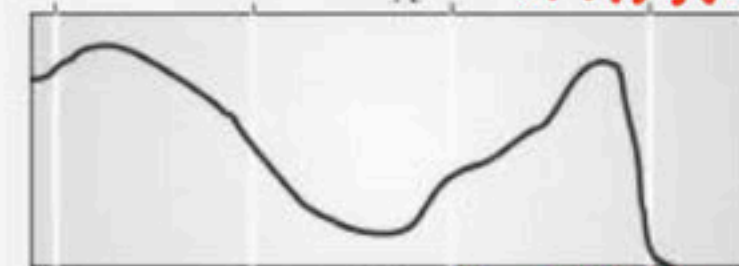


17. The following diagram shows different types of graph (A-C). Identify the graph showing action spectrum of photosynthesis?

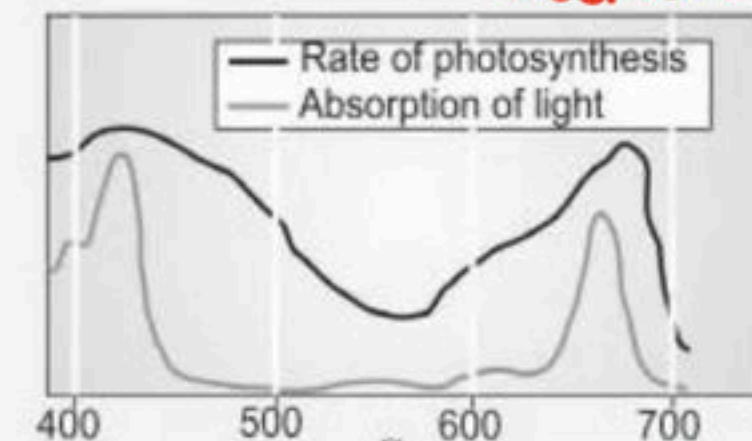
- ✓ (1) B only
- (2) A only
- (3) A, B only
- (4) B, C only



A → Absorption



B → action spectrum



C
→
Superimr.

18. Chlorophyll-a is ____ in colour

(1) Yellow green

✓ (2) Blue green

(3) Red green

(4) Yellow orange



19. During the dark reaction, the acceptor of CO_2 is

(1) NADPH_2

✓ (2) RuBP ✓

(3) H_2O

(4) CO_2



20. Statement A : C₄ photosynthetic pathway is more efficient than the C₃ pathway.

Statement B : Photorespiration is "suppressed" in C₄ plants.

- ✓ (1) Both statements are true and explaining each other.
- (2) Both statements are true but not explaining each other.
- (3) Only statement A is correct but statement B is not explaining the statement A .
- (4) Both statements are wrong.



21. Read the following statements, identify the correct statement

(1) Movement of electrons in ETS is coupled to pumping of protons into the lumen. ✓

(2) A hydrogen-carrier contributes in creation of proton gradient. ✓

(3) F₀ part of ATPase is associated with breakdown of proton gradient.

✓ (4) All are correct



22 Statement A : The plants in which first product of CO_2 fixation is a C_3 acid (OAA)
is called C_3 pathway and the plants in which first product was a C_4 acid (PGA) is
called C_4 pathway X

Statement B : Calvin pathway occurs in most of the photosynthetic plants X

- (1) Statement-A is correct but Statement-B is incorrect
- (2) Statement-A is incorrect but Statement-B is correct
- (3) Both the statements are correct
- (4) Both the statements are incorrect



23. During light reaction of photosynthesis. :-

- ✓ (1) Photolysis of water and photophosphorylation occurs
- (2) Photolysis of water and oxidative phosphorylation occurs ✗
- (3) No photolysis of water and photophosphorylation occurs
- (4) No photolysis of water and oxidative phosphorylation occurs



24. During photorespiration instead of being converted to
2 molecules of PGA RuBP forms_

- (1) One molecule of phosphoglycerate and phosphoglycolate
- (2) One molecule of phosphoenol^x and phosphoglycerate
- (3) One molecule of ethanol and pyruvic acid
- (4) One molecule of succinate and ethanol



25. Match the following Columns I & II and choose the correct option.

Column I		Column II	
A	Calvin cycle	P	ATP, NADPH, O_2
B	Cyclic photo-phosphorylation	Q	All photosynthetic plants
C	Non-cyclic photo-phosphorylation	R	Stroma lamellae

✓ (1) A-Q, B-R, C-P

(2) A-P, B-Q, C-R

(3) A-Q, B-P, C-R

(4) A-R, B-Q, C-P



26. Which of the following is the major pigment responsible for trapping light? →

(1) Xanthophyll

✓ (2) Chlorophyll

(3) Carotenoid

(4) All of these



27 Accessory pigments also absorb light and transfer the energy to _

(1) Chlorophyll-b

✓ (2) Chlorophyll-a

(3) Chlorophyll-c

(4) More than one option is correct



28 The binding CO₂ and O₂ with RuBisCO is ____.

(1) Symbiotic

✓(2) Competitive

(3) Co-operative

(4) None of the above



29. The large cells around the vascular bundles of the C_4 pathway plants are called?

(1) Mesophyll cells

✓ (2) Bundle sheath cells

(3) Passage cells

(4) Both 1 and 2



30. Unidirectional flow of electrons in
photophosphorylation takes place in

(1) Cyclic

✓ (2) Non-cyclic

(3) Both

(4) None



31 Dark reaction in photosynthesis is called so because

- (1) It can occur in dark also
- ✓ (2) It does not depend on light energy
- (3) It cannot occur during day light
- (4) It occurs more rapidly at night



32. Who demonstrated that O₂ comes from water instead of CO₂ during photosynthesis in their experiment, Also identify the organism used

- ✓ (1) Cornelius van Niel, purple and green bacteria
- (2) Engelmann van Niel, purple and green bacteria
- (3) Julius van Niel, green bacteria
- (4) Cornelius van Niel, green bacteria



33. Match the following Columns I & II and choose the correct option.

Column I		Column II	
A	Chemiosmotic mechanism	P	Phospho-enol pyruvic acid
B	CO ₂ acceptor in CAM plants	Q	Hill reaction
C	Light reaction	R	ATP synthesis

(1) A-P , B-Q , C-R

✓ (2) A-R , B-P , C-Q

(3) A-R , B-Q , C-P

(4) A-Q , B-P , C-R



34. Statement A : Chlorophyll a is bright or blue green and Chlorophyll b is yellow green ✓

→ carotenoids

Statement B : Xanthophyll is yellow-to yellow orange and carotenoids is yellow in colour ✗

- ✓(1) Statement-A is correct but Statement-B is incorrect
- (2) Statement-A is incorrect but Statement-B is correct
- (3) Both the statements are correct
- (4) Both the statements are incorrect



35. Statement A : Carbon dioxide is the major limiting factor for photosynthesis ✓

Statement B : C₄ plants show saturation at about ~~450~~³⁶⁰ μL^{-1} while C₃ plants show saturation beyond ~~360~~⁴⁵⁰ μL^{-1} ✗

- ✓ (1) Statement-A is correct but Statement-B is incorrect
- (2) Statement-A is incorrect but Statement-B is correct
- (3) Both the statements are correct
- (4) Both the statements are incorrect



36. Greenhouse crops such as tomatoes and bell pepper produce higher yields. This is due to:

- (1) CO_2 is a limiting factor to photosynthesis
- (2) Tomatoes and bell pepper are not C_3 plants
- ✓(3) CO_2 enriched atmosphere leads to higher yields
- (4) Due to diffused light in green house



37. The initial enzyme of Calvin cycle is

- ✓ (1) Ribulose 1,5 diphosphate carboxylase
- (2) Ribulose 1~~3~~ diphosphate carboxylase
- (3) Hexokinase
- (4) Cytochrome oxidase



38) During light reaction

(1) pH of stroma ↓es

(2) pH of lumen ↑es

✓ (3) pH of lumen ↓es

(4) pH of stroma remains unaffected



39. Which factor affects the photosynthesis?

- (1) Light quality ✓
- (2) Light intensity ✓
- (3) The duration of exposure to light ✓
- ✓ (4) All of the above



40. Statement A : A first action spectrum of photosynthesis resembles roughly the absorption spectra of chlorophyll a and b ✓

Statement B : Engelmann observed that the bacteria accumulated mainly in the region of ~~violet~~ ^{Blue} and red light of the split spectrum ✗

- ✓ (1) Statement-A is correct but Statement-B is incorrect
- (2) Statement-A is incorrect but Statement-B is correct
- (3) Both the statements are correct
- (4) Both the statements are incorrect



41. Match the following Columns I & II and choose the correct option.

Column I

A. Peter Mitchell

B. Melvin Calvin

C. Hatch and Slack

D. Blackman

(1) A-Q , B-S , C-R , D-P

(3) A-P , B-R , C-Q , D-S

Column II

^{C₄}
P. cycle

Q. Chemiosmotic hypothesis

R. Law of limiting factors

S. Dark reaction

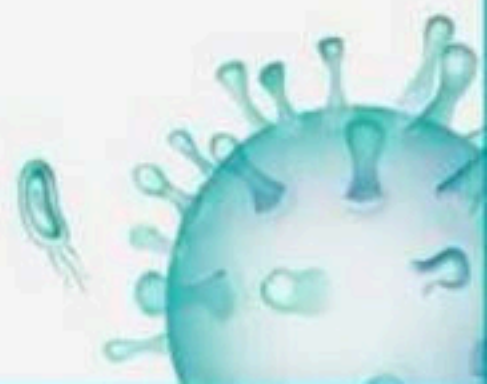
☒ (2) A-Q , B-S , C-P , D-R

(4) A-R , B-P , C-Q , D-S



42. In C₃ plants, the first stable product of photosynthesis during dark reaction is

- ✓ (1) 3-phosphoglyceric acid
- (2) Phosphoglyceraldehyde
- (3) Malic acid
- (4) Oxaloacetic acid



43. Identify the molecule in which the Photosystem I passes an electron to NADP^+ during cyclic photophosphorylation?

- (1) Cytochrome
- (2) Plastoquinone
- (3) Plastocyanin
- ✓ (4) Ferredoxin



44. Photophosphorylation and oxidative phosphorylation occurs in respectively

(1) Mitochondria, Mitochondria

✓ (2) Chloroplast, Mitochondria

(3) Cytoplasm, Chloroplast

(4) All of these



45. Assertion : To reduce the effect of photorespiration ~~C₃~~ C₄
plants have been gifted with CO₂ concentrating mechanism x

Reason : The CO₂ concentrating mechanism is called ~~C₃~~ pathway x

(1) Both Assertion and Reason are true and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

✓ (4) Both Assertion and Reason are false.



46. Statement A : Biosynthetic phase of photosynthesis directly depend on the presence of light but is independent on the products of the light reaction ^x

Statement B : The use of radioactive ^{14}C by Calvin in algal photosynthesis studies led to the discovery that the first CO_2 fixation product was a 3-carbon organic acid ✓

(1) Statement-A is correct but Statement-B is incorrect

✓ (2) Statement-A is incorrect but Statement-B is correct

(3) Both the statements are correct

(4) Both the statements are incorrect



47. The conversion NADP⁺ to H⁺ is an example of

- (1) Photosynthesis
- (2) Photolysis
- ✓ (3) Reduction
- (4) Oxidation



48. Select an incorrect statement for C₃ cycle :-

- (1) The regeneration of RuBP from glyceraldehyde- 3-
phosphate requires ¹~~2~~ ATP molecule each $\times$
- (2) With the oxidation of PGA, glyceraldehyde 3-
phosphate is formed.
- (3) C₃ cycle occurs only in C₃ plants $\times$
 $\rightarrow$ common
- (4) All are incorrect



49. Which range of wavelength (in nm) is called photosynthetically active radiation (PAR)?

(1) 100 – 390

(2) 390 – 430

✓ (3) 400 – 700 ✓

(4) 760 – 10,000



50 Free energy released in cyclic photophosphorylation_____

- (1) By the formation of ATP.
- (2) During the excitation of chlorophyll
- (3) Movement of protons in redox reactions of the electron transport chain ✗
- ✓ (4) Movement of electrons in redox reactions of the electron transport chain



51. Who proved that light is essential for plant process that purifies foul air by placing Priestly setup once in light and once in dark?

- (1) Jan Ingenhousz
- (2) Joseph Priestley
- (3) Von Sachs
- (4) Cornelius Van Niel



52. Match the following Columns I & II and choose the correct option.

Column I

A Chlorophyll-a

B Xanthophylls

C Chlorophyll-b

D Carotenoids

Column II

P Yellow-green

Q Yellow

R Yellow to Yellow orange

S Blue-green

(1) A-S , B-Q , C-R , D-P

(3) A-P , B-Q , C-R , D-S

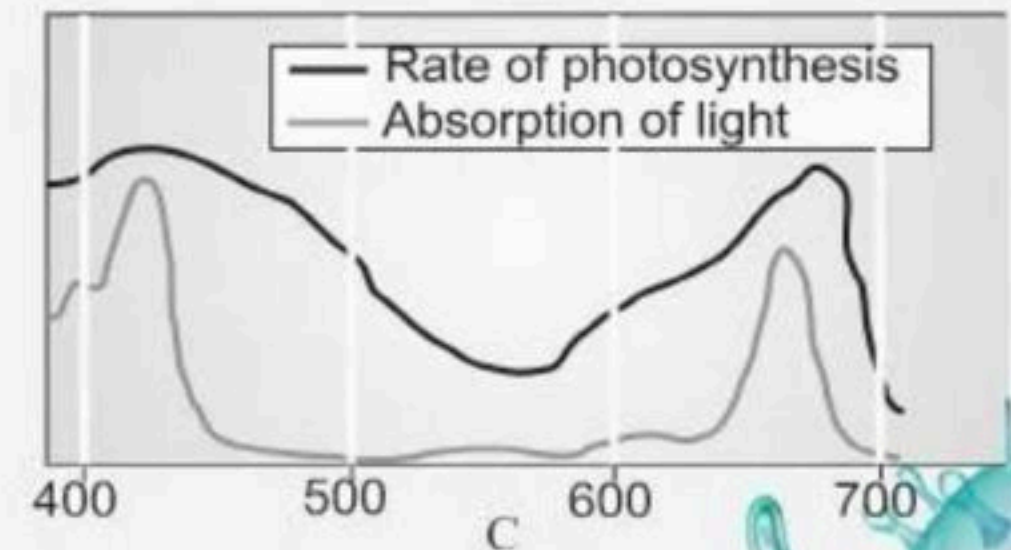
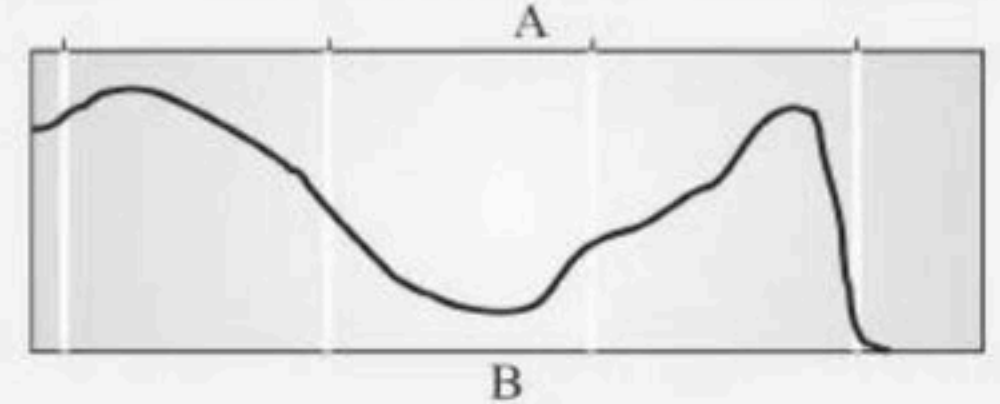
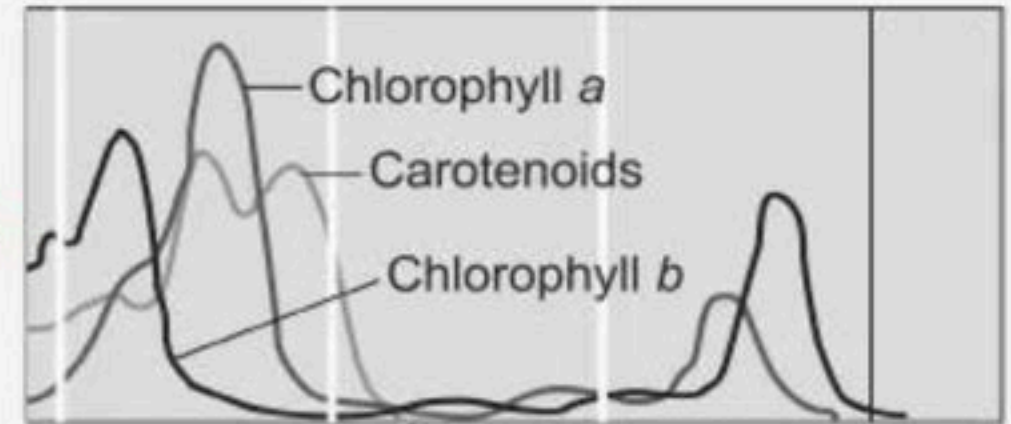
(2) A-S , B-R , C-Q , D-P

✓ (4) A-S , B-Q , C-P , D-R



53. Following graph shows action spectrum and absorption spectrum. Identify in which regions of the spectrum, most of the photosynthesis take place?

- (1) Blue light only
- (2) Red light only
- ✓ (3) Both 1 and 2
- (4) None of the above



54) Which factor is the factor which directly affects the process if its quantity is changed according to Blackman's law?

- (1) The factor with average value
- ✓ (2) The factor with its minimal value
- (3) The factor with its maximum value
- (4) The factor other than minimal factor

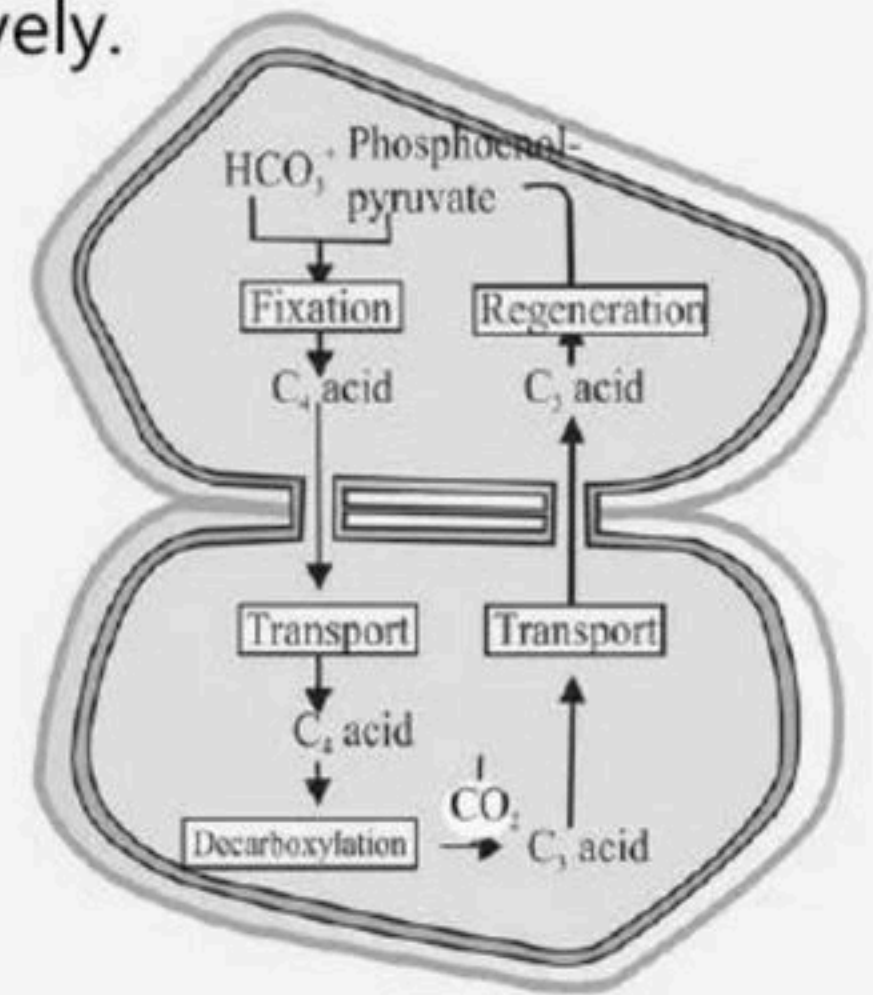


55) The given diagram represents the Hatch and Slack pathway. Name the CO_2 acceptor (A) and first stable product (B) of C_4 cycle respectively.

- (1) A - Phosphoglyceric acid, B - Phosphoenol pyruvate
- (2) A - RuBisCO, B - Phosphoglyceric acid
- (3) A - RuBisCO, B - Oxaloacetic acid
- (4) A - Phosphoenol pyruvate, B - Oxaloacetic acid

PEP

OAA



56 Which metal ion is a constituent of chlorophyll?

(1) Iron

(2) Copper

✓ (3) Magnesium

(4) Zinc



57. Why photorespiration is absent in C₄ plants?

- (1) Stomata closes when O₂ concentration is high
- (2) Stomata opens only when CO₂ concentration is high
- ✓ (3) Intracellular CO₂ concentration is high in bundle sheath and it ensures RuBisCO function as carboxylase
- (4) Intracellular CO₂ concentration is high in bundle sheath and it ensures RuBisCO function as oxygenase ⇒ photoresp.



58 Statement A : Using a prism, T.W Engelmann split light into its spectral components and then illuminated a green alga, Chlamydomonas, placed in a suspension of aerobic bacteria X

Statement B : The aerobic bacteria were used to detect the site of ~~O₂~~^{O₂} evolution X

(1) Statement-A is correct but Statement-B is incorrect

(2) Statement-A is incorrect but Statement-B is correct

(3) Both the statements are correct

✓ (4) Both the statements are incorrect



59. Cooperation of the two photosystems of the chloroplast is required for :-

- (1) ATP synthesis.
- ✓ (2) Reduction of NADP^+
- (3) Enhancement of dark reaction.
- (4) Generation of proton motive force ✗



60 Why productivity and yields are better in C⁴ plants?

- (1) It lacks photorespiration and don't tolerate high temperature
- (2) It shows photorespiration and can't tolerate high temperature
- ✓ (3) It lacks photorespiration and shows tolerance to high temperature
- (4) It shows photorespiration and tolerates high temperature.



61. Statement A : The LHC are made up of hundreds of pigment molecules bound to lipids ~~protein~~. x

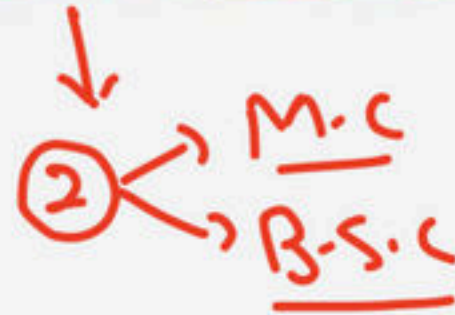
Statement B : Each photosystems has all the pigments (except chlorophyll-a) forming a light harvesting system also called antennae ✓

- (1) Statement-A is correct but Statement-B is incorrect
- ✓ (2) Statement-A is incorrect but Statement-B is correct
- (3) Both the statements are correct
- (4) Both the statements are incorrect



62. Total number of carboxylation in C₄ pathway is?

- (1) One
- (2) Three
- ✓ (3) Two
- (4) Four



63. At which percentage of the full sunlight light saturation occurs ?

- ✓ (1) 10 per cent
- (2) Less than 2%
- (3) Less than 5%
- (4) 0.15



64. The first stable product of CO_2 fixation in sorghum is:

(1) Pyruvic acid

✓ (2) Oxaloacetic acid

(3) Succinic acid

(4) Phosphoglyceric acid



65. How many components listed below are part of noncyclic ETS?

PS-II, NADP reductase, Hydrogen carrier, PS I, Water

Splitting complex, electron carrier

(1) Four

✓ (2) Three

(3) Five

(4) All are correct



66. What is correct about the C₄ plants?

→ Calvin pathway

- ✓ (1) Calvin pathway takes place in bundle sheath cells only
- (2) Calvin pathway takes place in mesophyll cells only ✗
- (3) Calvin pathway takes place in mesophyll as well as bundle sheath cells
- (4) None of these



67. Given below are some characteristics of photosynthetic plants. Which of these are true for C₄ plants :- PEP

- i) RuBisCo is the primary and first acceptor of CO₂ ✓
- ii) The first stable compound formed is 4-carbon compound ✓
- iii) The first stable compound formed is PGA × OAA
- iv) Possess PEP carboxylase ✓
- v) The first carboxylation reaction occurs in bundle sheath cells × M.C
- vi) The HSK cycle takes place in mesophyll cells ✓
- vii) 3-carbon in the primary CO₂ acceptor

(1) (i), (iv), (v), (vii)

(2) (i), (ii), (iii), (iv), (vii)

(3) (ii), (iii), (iv), (vi), (vii)

✓ (4) (ii), (iv), (vi), (vii)



68. Consider the following statements with respect to photosynthesis and identify the correct statements.

I. The first carbon dioxide acceptor in C₄ cycle is ~~PEP~~.

II. In C₃ plants, the first stable product of photosynthesis during dark reaction is ~~RuBP~~. PGA

III. Cyclic photophosphorylation results in the formation of ATP. ✓

IV. Oxygen which is liberated during photosynthesis comes from water. ✓

(1) I and II

(2) I and III

✓ (3) III and IV

(4) II and III



69. Match the following Columns I & II and choose the correct option.

Column I		Column II	
A	Dark reactions	P	CO_2 fixation in succulents
B	Photorespiration	Q	Two different types of chloroplasts
C	Hatch Slack Pathway	R	Carbondioxide is used
D	CAM Pathway	S	C_2 glycolate pathway

(1) A-Q , B-P , C-R , D-S

(2) A-R , B-Q , C-S , D-P

✓ (3) A-R , B-S , C-Q , D-P

(4) A-P , B-S , C-Q , D-R



70. Which of the following substance is used in half leaf experiment to absorb CO_2 ?

(1) KSO_4

✓ (2) KOH

(3) HNO_3

(4) Starch



71. During photosynthesis :-

I. PGA is changed into phosphoglyceraldehyde :- Reduction

II. NADPH is converted to NADP⁺ :- oxidation

Which of the following reaction occurs?

- (1) Oxidation, Reduction
- (2) Reduction, Reduction
- (3) Reduction, Electrolysis
- ✓ (4) Reduction, Oxidation



72. Statement A : RuBisCO has a much greater affinity for ~~O₂~~^{O₂}
than for ~~CO₂~~^{O₂}

Statement B : In C₃ plants, some O₂ binds with RUBP to
form one molecule of phosphoglycerate and
phosphoglycolate in a pathway called photorespiration

- (1) Statement-A is correct but Statement-B is incorrect
- ✓(2) Statement-A is incorrect but Statement-B is correct
- (3) Both the statements are correct
- (4) Both the statements are incorrect



73. Match the following Columns I & II and choose the correct option.

Column I		Column II	
A	PEPcase	P	Bundle sheath cells of C_4 plants
B	RuBisCO	Q	ATP and $NADPH_2$
C	Assimilatory power	R	Mesophyll cells of C_4 plants

(1) A-P , B-R , C-Q

✓ (2) A-R , B-P , C-Q

(3) A-Q , B-P , C-R

(4) A-R , B-Q , C-P



74 In Sorghum, which set of cells respectively represent C₄ and C₃ pathways occurring in them?

- (1) Bundle sheath, mesophyll cell
- (2) Epidermal cell, mesophyll cell
- (3) Stomatal guard cell, Bundle sheath
- (4) Mesophyll cell, Bundle sheath



75 Match the following Columns I & II and choose the correct option.

Column I		Column II	
A	Light reactions	P	Carbohydrate is formed
B	Chlorophyll-a	Q	<u>ATP</u> & <u>NADPH</u> synthesized
C	Dark reactions	R	$C_{40}H_{56}O_2$
D	Xanthophyll	S	$C_{55}H_{72}O_5N_4$ <u>Mg</u>

✓ (1) A-Q, B-S, C-P, D-R

(2) A-Q, B-R, C-P, D-S

(3) A-R, B-S, C-P, D-Q

(4) A-P, B-S, C-Q, D-R



76. Match the columns I and II.

Column I		Column II	
A	Growth at a <u>cellular</u> level	P	17,500 new cells per hour
B	Maize root apical meristem	Q	Increase in size by upto 3,50,000 times
C	Watermelon cells	R	Increase in the <u>amount</u> of protoplasm
D	Growth of a pollen tube	S	Measurement in terms of surface area
E	Growth in a dorsiventral leaf	T	Measurement in terms of length

- (1) A – R, B – P, C – Q, D – T, E – S
- (2) A – Q, B – R, C – P, D – T, E – S
- (3) A – S, B – R, C – P, D – T, E – Q
- (4) A – Q, B – T, C – P, D – R, E – S



77. All cells of a plant are descendants of which of the following?

- ☒ (1) Zygote
- (2) Fruits
- (3) Leaves
- (4) Stem



78. Read the following statements carefully and select the correct number of incorrect statements.

- (a) Trees continue to increase in height or girth over a period of time. ✓
- (b) There is relationship between the structure of a cell, a tissue, an organ and the function they perform. ✓
- (c) Descendants of zygote have different structural and functional attributes

(1) a only

(2) b only

(3) a and b only

✓ (4) All 3 are correct



79. Statement A : The leaves, flowers and fruits of the same tree have limited dimensions but also appear and fall periodically and some time repeatedly.

Statement B : All plant organs are made up of a variety of tissues.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- ☒ (3) Both statements are correct.
- (4) Both Statements are incorrect.



80. With respect to which of the following growth can be defined as an irreversible permanent increase in size?

- (1) An organ ✓
- (2) Parts of an organ ✓
- (3) An individual cell ✓
- (4) All of the above ✓



81. Plant growth is unique which statement supports this statement?

- (1) The plant retains the regeneration power. ✗
- ✓ (2) Capacity of the plants of unlimited growth
- (3) There is change in offspring's characters.
- (4) None of the above



82. Growth is regarded as one of the most____
characteristics of a living being.

- (1) Fundamental and inconspicuous
- ✓(2) Fundamental and conspicuous
- (3) Superfluous and conspicuous
- (4) Secondary and inconspicuous



83. ^x Statement A : Growth is accompanied by metabolic ^{+ catab.} process (Only anabolic), that occur at the expense of energy.

Statement B : Growth can be defined as an irreversible [✓] permanent increase in size of an organ or its parts or even of an individual cell

(1) Statement A is correct but Statement B is incorrect.

[✓] (2) Statement A is incorrect but Statement B is correct.

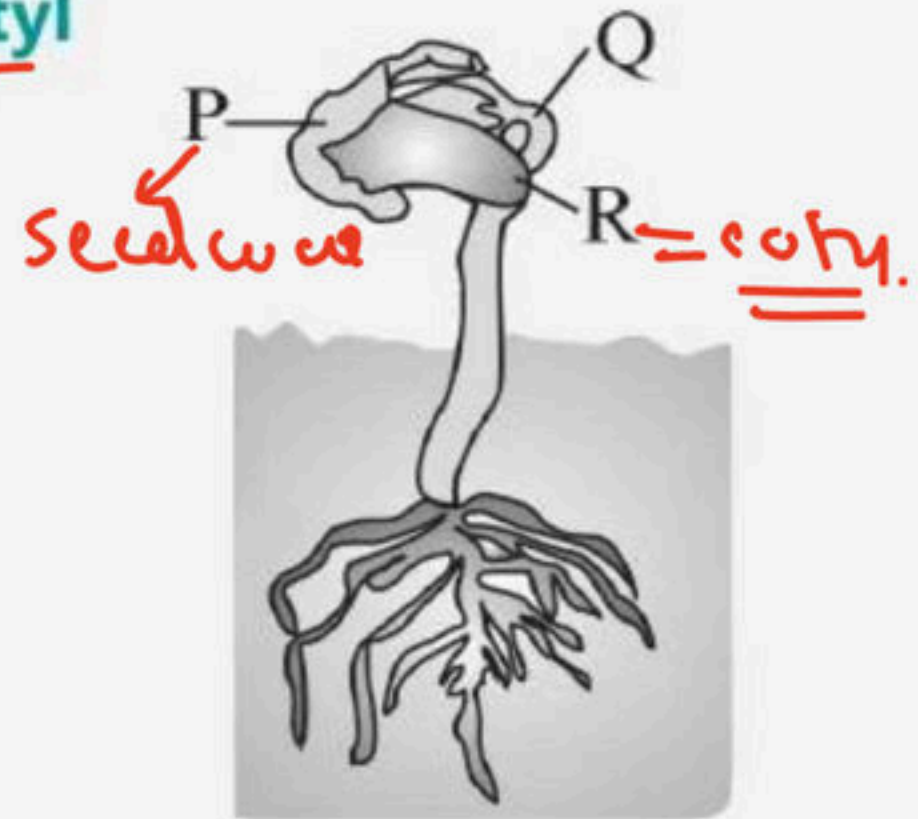
(3) Both statements are correct.

(4) Both Statements are incorrect.



84. Which of the following label represents 'epicotyl
hook'_

- (1) P
- ✓ (2) Q
- (3) R
- (4) None of the above



85. Which of the following statement is correct w.r.t growth in plants?

I. Plants retain the capacity for unlimited growth throughout their life

II. Meristem cells in plants have the capacity to divide and self-perpetuate ✓

III. Presence of meristems at certain locations in their body enable plants to grow

IV. New cells are always being added is open form of growth ✓

(1) I, II & III

(2) Both II & III

✓ (3) I, II, III & IV

(4) Both III & IV



86. Assertion : As whole plant growth is indefinite. ✓

Reason : Plants retain the capacity of continuous growth throughout their life.

- ✓ (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect.



87. The cell wall of the cells in meristematic region are ____

- (1) Primary in nature ✓
- (2) Thin and cellulosic ✓
- (3) With abundant plasmodesmatal connections ✓
- (4) ✓ All of the above



88. Match the columns I and II.

Column I		Column II	
A	Meristematic phase	P	Increased vacuolation
B	Elongation phase	Q	Constantly <u>dividing</u> cells
C	Maturation phase	R	Maximal size in terms of wall thickening

(1) A - R, B - P, C - Q

(2) A - Q, B - R, C - P

(3) A - P, B - R, C - Q

☒ (4) A - Q, B - P, C - R



89. Which of the following phase of growth represents the phase with constantly dividing cells, both at the root apex and the shoot apex?

- (1) The elongation phase
- (2) The maturation phase
- ✓ (3) The meristematic phase
- (4) The degradation phase



90. Identify the correct statements about meristematic phase of growth. :-

I. It is characterised by increased vacuolation. X

II. It shows cell enlargement. X

III. In this phase, new cell wall is deposited. X

IV. Growth is maximum. X

(1) III, I and II

(2) I, II and III

(3) II, IV and III

☒ (4) All are wrong



91. Match the columns I and II.

Column I		Column II	
A	Meristematic phase	P	Proximal to the <u>phase of elongation</u>
B	Elongation phase	Q	<u>Root/shoot apex</u>
C	Maturation phase	R	Proximal to <u>meristematic zone</u>

(1) A – R, B – P, C – Q

✓ (2) A – Q, B – R, C – P

(3) A – P, B – R, C – Q

(4) A – Q, B – P, C – R



92 Root apical meristem and shoot apical meristem
principally contribute to the elongation of the plants ____

(1) Along the radius

✓ (2) Along their axis

(3) Along diameter

(4) Along tangent



93. Statement A : Growth, at a cellular level, is principally a consequence of increase in the amount of protoplasm. ✓✓

Statement B : One single maize root apical meristem can give rise to more than ^{17,500} ~~3,50,000~~ new cells per hour ✗

- ✓ (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



94. Which of the following two processes can be summed as development.

(1) Wilting and swelling

✓ (2) Growth and differentiation

(3) Wilting and differentiation

(4) Growth and imbibition



95. Cell / Tissue arising out of the same meristem have different structures at maturity the phenomena in plants is called :-

- (1) Open growth
- ✓ (2) Open differentiation
- (3) Indeterminate growth
- (4) Determinate differentiation



96. Statement A : The period of growth is generally divided into three phases, namely, meristematic, elongation and maturat^{Plants}ion, in animals. X

Statement B : The cells in this region are rich in protoplasm, possess ~~X~~ inconspicuous nuclei. X

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- ☒ (4) Both Statements are incorrect.



97. Select the correct character related to the cell found in phase of elongation. :-

- (1) Cell wall is thin and cellulosic.
- (2) Abundant plasmodesmatal connections.
- ☒ (3) Increased vacuolation of cell
- (4) Cell wall becomes very much thickened.



98.) Read the following statements about the :-

I. Phase of maturation-Cells attain their maximal size, proper wall thickening and protoplasmic modifications, the cells in this region are rich in protoplasm, possess large conspicuous nucleus. X

II. Phase of elongation- It is characterised by rapid cell division, the cells proximal (just next, away from the tip) to the meristematic zone X

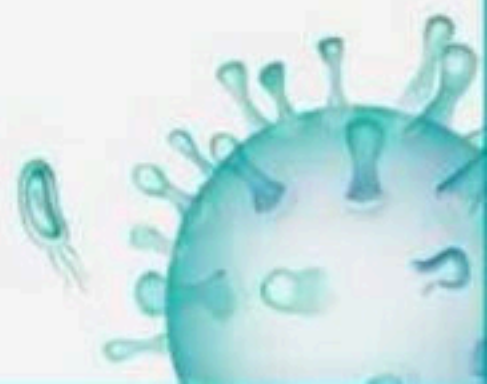
Choose the correct option.

(1) I and II

(2) II only

(3) I only

✓ (4) Both are wrong



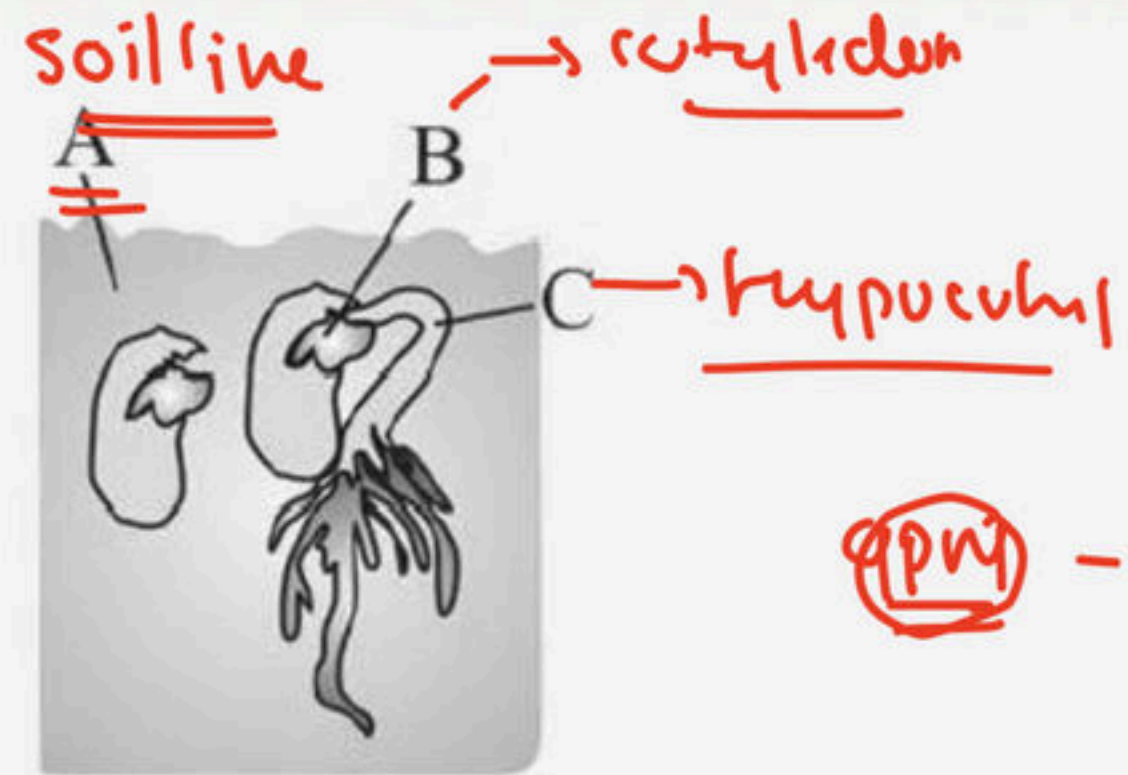
99. Growth is accompanied ____

- (1) By metabolic processes (both anabolic and catabolic)
- (2) At expense of energy
- ✓ (3) Both 1 and 2
- (4) By only anabolic processes



100. Label the following diagram of seedling_

- (1) A - Soil line, B - Cotyledon, C - Hypocotyl
 (2) A - Soil line, B - Cotyledon, C - Epicotyl
 (3) A - Rock line, B - Hypocotyl, C - Root Axis
 (4) A - Soil line, B - Hypocotyl, C - Cotyledon



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101. How many of the following are the effects influenced by ethylene?

- 1) Synchronising fruit set in pineapples**
- 2) Xylem differentiation**
- 3) Promotes female flowers in cucumbers**
- 4) Produce new leaves and chloroplasts in leaves**
- 5) Bolting in cabbage**

(1) 1, 3, 5

(2) 1, 3

(3) 1, 2, 4

(4) 2, 3, 4



102. Select the correct statements regarding auxins.

I. Auxins promote flowering e.g. in pineapple and promote abscission of new leaves and fruits.

II. Auxin controls dicot weed

- (1) I and II
- (2) II is correct
- (3) I is correct
- (4) Both are wrong



103. Auxins promote flowering in

- (1) Tomatoes
- (2) Pineapples
- (3) Oranges
- (4) Apples



104. Statement A : Auxin controls phloem differentiation and helps in cell division.

Statement B : There are more than 100 gibberellins reported from widely different organisms such as fungi and higher plants.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



105. Statement A : Cytokinins were discovered as kinetin (a modified form of adenine, a purine) from the autoclaved herring sperm DNA.

Statement B : Kinetin does not occur naturally in plants.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



106. Bacterial pathogen responsible for foolish seedling disease of rice seedlings is

- (1) *Gibberella kurosawa*.
- (2) *Gibberella fujikuroi*.
- (3) *Gibberella moniliformae*.
- (4) None of them



107. Cytokinins help to produce

- (1) New leaves and chloroplast in leaves
- (2) Lateral shoot
- (3) Adventitious shoot
- (4) All of them



108. Statement A : Difference in shapes of leaves produced in air and those produced in water in buttercup also represent the heterophyllous development due to phase of life.

Statement B : Development in plants (i.e., both growth and differentiation) is under the control of intrinsic and extrinsic factors.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



109. Match the columns I and II.

Column I		Column II	
A	Auxin	P	Flowering in pineapples
B	Gibberellin	Q	Rapid cell division
C	Cytokinins	R	Delay senescence
D	Ethylene	S	Fastens ripening

- (1) A – R, B – P, C – Q, D – S
(2) A – S, B – R, C – P, D – Q
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S



110. Identify the mismatch

- (1) Promotion of flowering in pineapple Ethylene
- (2) Seed dormancy - ABA
- (3) Apical dominance - Auxins
- (4) Senescence in leaves - Ethylene



111. Identify the correct statement

- I. Natural cytokinins are synthesised in regions where rapid cell division occurs**
- II. In tissue culture both auxin and cytokinins are used for growth and differentiation.**
- III. In callus, root regeneration is promoted by auxin.**
- IV. Cytokinins play a vital role in morphogenesis in plants.**

(1) I, II and III

(2) Only IV

(3) III and IV



(4) All are correct

112. Why is ABA called stress hormone?

(1) Removes various stresses from plant

(2) Increase tolerance of plant to various kind of stresses

(3) Both (1) and (2)

(4) None of these



113. Apples are generally wrapped in waxed paper to

- (1) Prevent sunlight for changing its colour
- (2) Prevent aerobic respiration by checking the entry of
- (3) Prevent ethylene formation due to injury
- (4) Make the apples look attractive



114. Statement A : Further away from the apex, i.e., more distal to the phase of elongation, lies the portion of axis which is undergoing the phase of maturation.

Statement B : The cells of maturation zone, attain their maximal size in terms of wall thickening and protoplasmic modifications.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



115. Statement A : Turgidity of cells inhibits extension growth.

Statement B : Oxygen helps in releasing metabolic energy essential for growth activities.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



116. What do you call an expansion of leaf?

- (1) Growth
- (2) Redifferentiation
- (3) Wilting
- (4) Centrifugation



117. Identify the incorrect statement/s

I. Kinetin is a derivative of Adenine which is a pyrimidine.

II. The technique of decapitation is widely used in tea plantations.

III. Ethylene is a gaseous plant hormone.

IV. Use of , hastens the malting process in brewing industry.

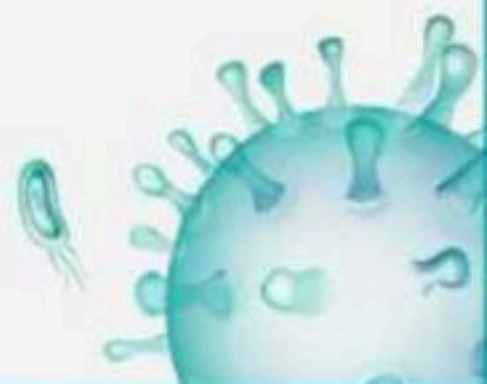
V. is a growth promoter.

(1) I,II,III,IV

(2) Only III

(3) II,III,IV

(4) I and V



118. Formation of interfascicular cambium from fully developed parenchyma cells is an example for

- (1) Differentiation
- (2) Redifferentiation
- (3) Dedifferentiation
- (4) Maturation



119. Identify the correct statement.

I. Elongation of a plant along the axis is called primary growth

II. Increase in girth of plants occurs due to the apical meristem.

III. Secondary growth of plants occurs due to the lateral meristem.

IV. Vascular cambium and cork cambium are the lateral meristems of plants.

(1) I, III and, IV

(2) III and IV

(3) IV and II

(4) I and IV



120. Which one includes growth promoters?

- (1) Auxin, cytokinin, ABA
- (2) Auxin, cytokinin, GA_3
- (3) C_2H_2 , ABA
- (4) GA, C_2H_4



121. The exponential growth can be expressed as

(1) $W_1 \text{ ert} = W_0$

(2) $L_t = L_0 + rt$

(3) $W_1 = W_0 e^{rt}$

(4) $L_t = L_0 + \text{ert}$



122. Statement A : The period of growth is generally divided into three phases, namely, meristematic, elongation and maturation, in animals.

Statement B : The cells in this region are rich in protoplasm, possess inconspicuous nuclei.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



123. Match the columns I and II.

Column I		Column II	
A	Heterophylly in cotton	P	Plasticity due to environment
B	Heterophylly in buttercup	Q	Sum of growth and differentiation
C	Development	R	Plasticity due to phase of life

- (1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R



124. Arrange the following events in correct sequence, in the life cycle of a typical dicot plant.

p) Differentiation

q) Redifferentiation.

r) Dedifferentiation

s) Divisions in primary meristem.

(1) p, q, r, s

(2) s, p, r, q

(3) s, r, p, q

(4) q, p, r, s



125. Statement A : Environmental signals such as light and gravity also affect certain phases/stages of growth.

Statement B : During differentiation, cells undergo few to major structural changes in their cell walls and not protoplasm.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



126. Which of the following is/are factor causing seed dormancy?

- (1) Impermeable and hard seed coat
- (2) Chemical inhibitors like abscisic acids, phenolic acids and para-ascorbic acids
- (3) Immature embryos
- (4) All of them



127. Which ability of gibberellins is used to increase the length of grape stalks?

- (1) Ability to promote lateral dominance
- (2) Ability to increase in length of axis
- (3) Ability to promote apical dominance
- (4) All of these



128. Which of the following parts of tree have limited dimensions, fall periodically and some time repeatedly?

- (1) The leaves, flowers and fruits
- (2) The stem, leaves and flowers
- (3) The stem, root and twigs
- (4) The leaves, fruits and roots



129. During water stress, rise in ABA levels initially cause

- (1) Stomatal opening
- (2) Stomatal closure
- (3) Increased transpiration
- (4) Reduced transpiration



130. Find out the

i. Gaseous hormone from the following.

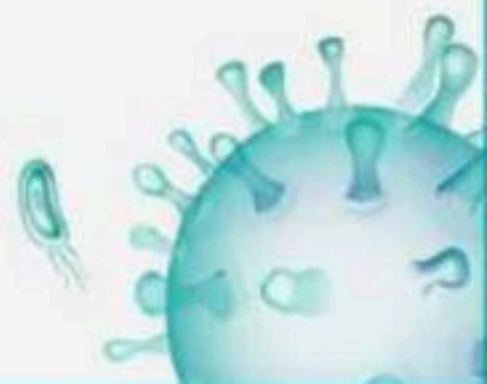
ii. Hormone that could fit either of the groups, but it is largely an inhibitor of growth activities

- (1) Ethylene, auxins
- (2) Ethylene, gibberellins
- (3) Cytokinins, ethylene
- (4) Ethylene, ethylene



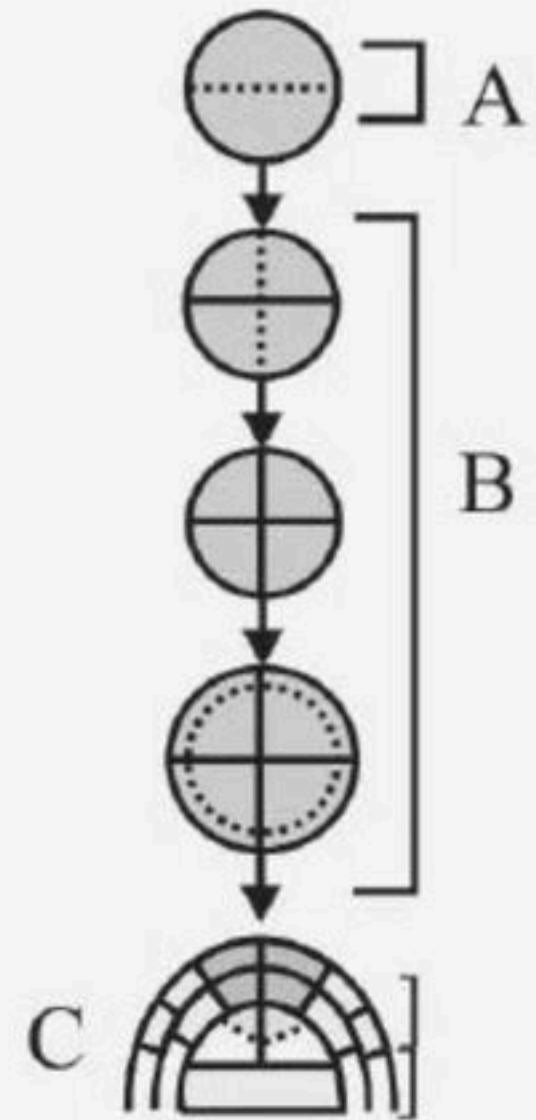
131. In arithmetic growth pattern, on plotting the length of the organ against time, ____.

- (1) A linear curve is obtained.
- (2) A parabolic curve is obtained
- (3) A hyperbolic curve is obtained
- (4) A sigmoid curve is obtained



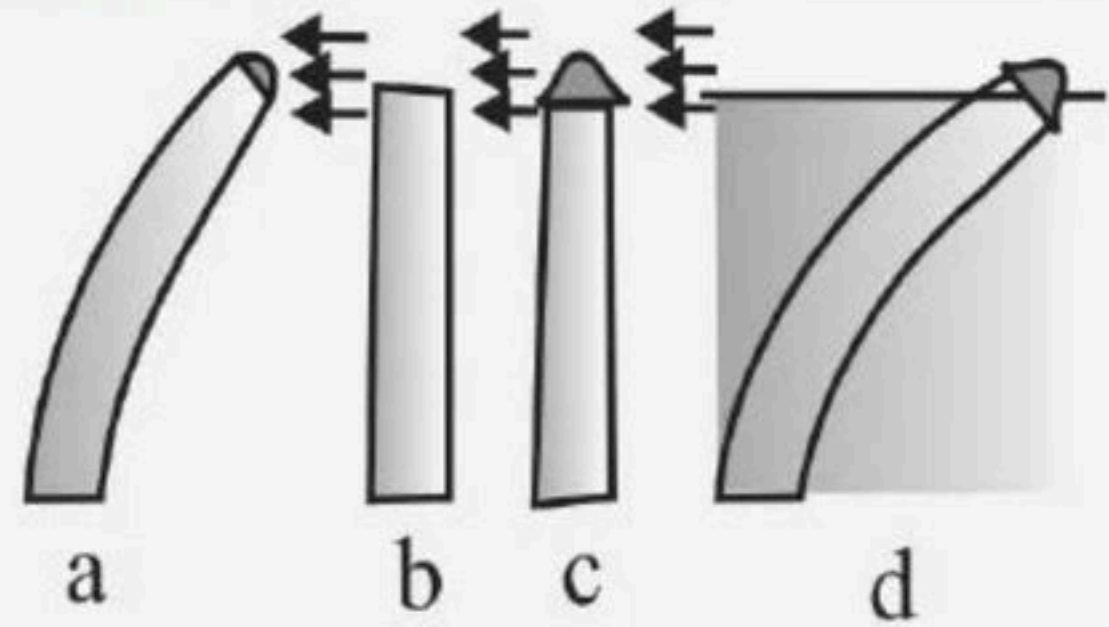
132. Label the given diagram

- (1) A- Zygote divided, B- Arithmetic phase, C- Geometric phase
- (2) A-Zygote divided, B- Geometric Phase, C-Arithmetic phase
- (3) A- Zygote maturation, B- Arithmetic phase, CGeometric phase
- (4) A- Zygote maturation, B- circulars phase, CGeometric phase



133. Who among the following conducted an experiment depicted by following diagram_____

- (1) F. Skoog and his co-workers
- (2) E. Kurosawa
- (3) Cousins
- (4) Charles Darwin and his son Francis Darwin



134. Assertion : The sum of growth and differentiation is development.

Reason : Development in plants is under the control of extrinsic factors only.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect.



135. All are types of roles played by PGR like

- (1) Individualistic or synergistic
- (2) Complimentary or antagonistic
- (3) 1 and 2
- (4) None of these



136. In most systems, the initial growth is _A_, and it B_ rapidly thereafter - at an _C_

- (1) A - slow (lag phase), B- exponential rate (log or exponential phase),
C- increases
- (2) A - slow (lag phase), B - decreases,
C- exponential rate (log or exponential phase)
- (3) A - slow (lag phase), B- increases,
C- exponential rate (log or exponential phase)
- (4) A - slow (log phase), B- increases,
C - exponential rate (lag or exponential phase)



137. Charles Darwin used which plant in the discovery of plant growth regulator?

- (1) Larkspur
- (2) Buttercup
- (3) Tobacco
- (4) Canary grass



138. Match the columns I and II.

Column I		Column II	
A	Extrinsic factor for growth	P	Plasticity
B	Intrinsic factors for growth	Q	Light, temperature, water
C	Heterophylly in cotton, coriander, larkspur, buttercup	R	Genetics and plant growth regulators

- (1) A – R, B – P, C – Q
(2) A – Q, B – R, C – P
(3) A – P, B – R, C – Q
(4) A – Q, B – P, C – R



139. Match the columns I and II.

Column I		Column II	
A	Initiate rooting in stem cuttings	P	Gibberellins
B	Bolting in beet cabbages and many plants with rosette habit.	Q	Auxin
C	Autoclaved herring sperm DNA	R	Kinetin

- (1) A – R, B – P, C – Q
- (2) A – Q, B – R, C – P
- (3) A – P, B – R, C – Q
- (4) A – Q, B – P, C – R



140. Match the following:

Column I		Column II	
A	IAA	P	Herring sperm DNA
B	ABA	Q	Bolting
C	Ethylene	R	Stomatal closure
D	GA	S	Weed-free lawns
E	Cytokinins	T	Ripening of fruits

- (1) A – S, B – R, C – T, D – Q, E – P
(2) A – T, B – R, C – S, D – Q, E – P
(3) A – S, B – P, C – T, D – R, E – Q
(4) A – T, B – R, C – Q, D – P, E – S



141. Match the columns I and II.

Column I		Column II	
A	Charles Darwin and Francis Darwin	P	rice seedlings
B	F. W. Went	Q	Volatile substance in ripened oranges
C	E. Kurosawa	R	Isolation of auxin from oat seedling
D	F. Skoog and his co-workers	S	Phototropism in canary grass
E	Cousins	T	Internodal segments of tobacco stems

- (1) A – R, B – P, C – Q, D – T, E – S
- (2) A – Q, B – R, C – P, D – T, E – S
- (3) A – S, B – R, C – P, D – T, E – Q
- (4) A – Q, B – T, C – P, D – R, E – S



142. Match the columns I and II.

Column I		Column II	
A	Indole compounds	P	IAA
B	Adenine derivatives	Q	Abscissic acid, ABA
C	Derivatives of carotenoids	R	N6 -furfurylamino purine, kinetin
D	Terpenes	S	Gibberellic acid, GA3

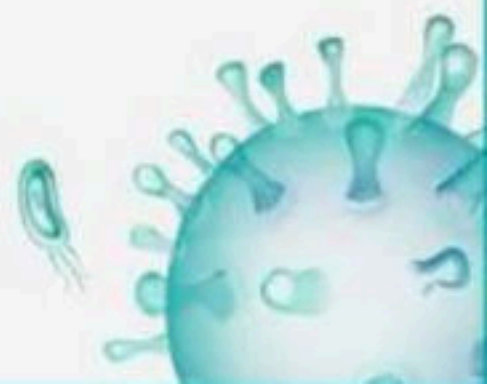
- (1) A – R, B – P, C – Q, D – S
(2) A – S, B – R, C – P, D – Q
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S



143. Match the columns I and II.

Column I		Column II	
A	Apical hook formation in dicot seedlings	P	Gibberellins
B	Nutrient mobilisation	Q	Abscissic acid
C	Extend the market period of fruits	R	Cytokinins
D	Inhibition of seed germination	S	Ethylene

- (1) A – R, B – P, C – Q, D – S
(2) A – S, B – R, C – P, D – Q
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S



144. Match the columns I and II.

Column I		Column II	
A	Auxin	P	Hastening the maturity period of conifers
B	Cytokinins	Q	Xylem differentiation
C	Gibberelins	R	Overcome the apical dominance
D	Ethylene	S	Tolerance of plants to various kinds of stresses
E	Abscissic acid	T	Breaks seed and bud dormancy

- (1) A – R, B – P, C – Q, D – T, E – S
- (2) A – Q, B – R, C – P, D – T, E – S
- (3) A – S, B – R, C – P, D – T, E – Q
- (4) A – Q, B – T, C – P, D – R, E – S



145. Match the entries in Column I with those of Column II and choose the correct answer.

Column I		Column II	
A	Cytokinin	P	Stress hormone
B	Auxins	Q	Ripening of fruits
C	Abscisic acid	R	Apical dominance
D	. Ethylene	S	Bolting
		T	.Richmond Lang effect

- (1) A - T, B - R, C - P, D - Q
- (2) A - T, B - R, C - Q, D - S
- (3) A - R, B - S, C - Q, D - P
- (4) A - Q, B - Q, C -T,D - R



146. Match the columns I and II.

Column I		Column II	
A	Arithmetic growth	P	$W_1 = W_0 e^{rt}$
B	Geometric growth	Q	$L_t = L_0 + rt$
C	Growth rate	R	r
D	Base of natural logarithms	S	e

- (1) A – R, B – P, C – Q, D – S
(2) A – S, B – R, C – P, D – Q
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S



147. Match the columns I and II.

Column I		Column II	
A	Cells of meristems	P	Primary growth of plant
B	New cells are always being added	Q	Secondary growth of the plant
C	Apical meristem	R	Open form of growth
D	Vascular cambium and cork-cambium	S	Divide and self-perpetuate

- (1) A – R, B – P, C – Q, D – S
- (2) A – S, B – R, C – P, D – Q
- (3) A – P, B – R, C – Q, D – S
- (4) A – Q, B – P, C – R, D – S



148. Identify the correct function of GA.

I. Acts as general plant growth inhibitor and an inhibitor of plant metabolism.

II. Plays an important role in seed development, maturation and dormancy

III. Helps seeds withstand desiccation and other factors unfavourable for growth.

(1) I

(2) I, II

(3) I, II, III

(4) None of them



149. Statement A : Gibberellins fastens senescence.

Statement B : GA3 is used to speed up the malting process in brewing industry.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



150. How does gibberellins change the quality of an apple?

- (1) Make it sweeter
- (2) Elongate it in size
- (3) Improve its shape
- (4) 2 and 3



151. What amount of energy is released from glucose during lactic acid fermentation

- (1) More than 18%
- (2) About
- (3) Less than
- (4) Approximately



152. RQ for carbohydrate is

- (1) 0.8
- (2) 0.7
- (3) 0.9
- (4) 1



153. Statement A : The continued oxidation of acetyl CoA via the TCA cycle requires the continued replenishment of citric acid

Statement B : In addition it also requires regeneration of NADH and from NAD⁺ and FAD⁺

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



154. When 8H^+ pass through $\text{F}_0 - \text{F}_1$ particle, how many ATPs are produced?

- (1) 6 ATP
- (2) 4 ATP
- (3) 8 ATP
- (4) 2ATP



155. Match Column I with Column II

Choose the correct answer from the options given below:

Column I		Column II	
A	Citric acid cycle	P	Cytoplasm
B	Glycolysis	Q	Mitochondrial matrix
C	Electron transport system	R	Intermembrane space of mitochondria
D	Proton gradient	S	Inner mitochondrial membrane

- (1) A – P, B – Q, C – R, D – S
- (2) A – Q, B – P, C – S, D – R
- (3) A – R, B – S, C – P, D – Q
- (4) A – S, B – R, C – Q, D – P



156. During aerobic breakdown of Glucose what is the series of intermediates formed with respect to no. of carbon atoms that they have

I. 3C compound

II. 4C compound

III. 5C compound

IV. 2C compound

V. 6C organic acid

(1) I, II, III, IV, V

(2) II, IV, V, I, II

(3) I, IV, V, III, II

(4) I, V, IV, III, II



157. To form a respiratory balance sheet, the calculations can be made on certain assumptions like

- (1) There is a sequential pathway functioning like glycolysis, TCA cycle and ETS following one after another.
- (2) NADH formed in glycolysis is transferred to mitochondria and undergoes oxidative phosphorylation.
- (3) None of the intermediates in the pathway are utilized to synthesize any other compound and glucose is being respired no other alternative substrate are entering in the pathway at any of the intermediary stages.
- (4) All of them



158. Oxidative decarboxylation occurs during the formation of:

- (1) Citric acid and Succinic acid
- (2) Citric acid and Oxaloacetic acid
- (3) Acetyl CoA and Succinyl CoA
- (4) Oxaloacetic acid and Oxalosuccinic acid



159. At how many places in Kreb's cycle $\text{NADH} + \text{H}^+$ is formed?

- (1) 1
- (2) 2
- (3) 3
- (4) 4



160. Partial oxidation of glucose into pyruvic acid is known as

- (1) Glycolysis
- (2) Kreb's cycle
- (3) ETS
- (4) All of these



161. Pyruvate dehydrogenase complex needed for the conversion of pyruvic acid to acetyl CoA is located in:

- (1) Intermembranal space of mitochondria
- (2) Matrix of mitochondria
- (3) Cytoplasm
- (4) Grana of chloroplast



162. Match the columns I & II and choose the correct option.

Column I		Column II	
A	End product of fermentation in yeast cells	P	2 molecules
B	Net ATP produced in fermentation	Q	$\text{NADH} + \text{H}^+$
C	End product of fermentation in muscle cells during exercise	R	Ethanol and CO_2
D	Reducing agent in respiration process	S	Lactic acid and energy

(1) A-P, B-Q, C-R, D-S

(2) A-P, B-R, C-S, D-Q

(3) A-R, B-P, C-S, D-Q

(4) A-Q, B-R, C-S, D-P



163. Arrange the steps of glycolysis in correct order

i. Conversion of BPGA to 3-Phosphoglyceric acid

ii. Conversion of 3-Phosphoglyceric acid to 2- Phosphoglycerate

**iii. Splitting of Fructose 1,6 -biphosphate into dihydroxy acetone phosphate and 3-
3-**

Phosphoglyceraldehyde

iv. Conversion of 3-Phosphoglyceraldehyde to 1,3- biphosphoglycerate

(1) iii, iv, i, ii

(2) iii,i, ii, iv

(3) i, iii, ii, iv

(4) iv, ii, iii, i



164. Match the columns I & II and choose the correct option.

Column I		Column II	
A	Cytochrome c	P	Ubiquinone
B	Complex IV	Q	ATP synthase
C	Complex V	R	Small protein attached to the outer surface of the inner mitochondrial membrane
D	Mobile electron carrier	S	Contains cytochromes a and a ₃ and 2 copper centres

(1) A-R, B-S, C-P, D-Q

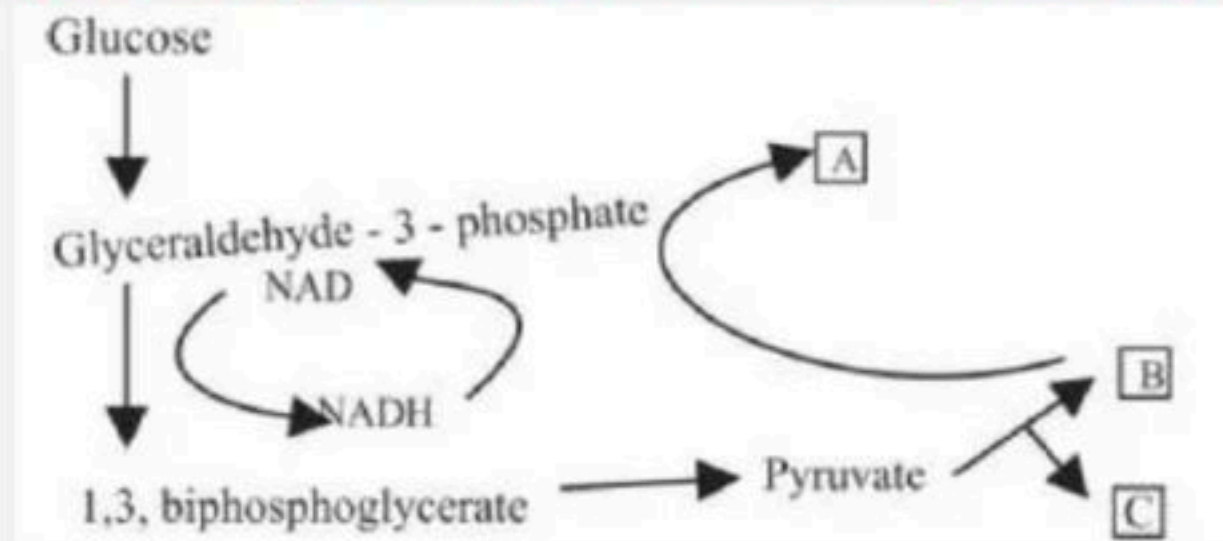
(2) A-R, B-S, C-Q, D-P

(3) A-Q, B-R, C-S, D-P

(4) A-Q, B-P, C-S, D-R



165. Choose the correct combination of labelling the molecules involved in the pathway of anaerobic respiration in Yeast.



- (1) A - Ethanol, B - CO₂, C - Acetaldehyde
- (2) A - CO₂, B - Ethanol, - Acetaldehyde
- (3) A - Acetaldehyde, B- CO₂, C- Ethanol
- (4) A - Ethanol, B - Acetaldehyde, C - CO₂



166. Statement A : When fats are used in respiration, the RQ is more than 1

Statement B : When proteins are used as respiratory substrates the ratio would be about 0.9

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



167. Yeast poisons themselves to death when alcohol concentration reaches to alcoholic fermentation.

- (1) 7%
- (2) 10%
- (3) 13%
- (4) 20%



168. Assertion : Conversion of 1,3 biphosphoglycerate to 3-phosphoglycerate is an energy yielding step

Reason : This energy is trapped by formation of ATP

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



169. In the electron transport system, the final acceptor of proton is

- (1) cytochrome b
- (2) cytochrome a_3
- (3) oxygen
- (4) ubiquinone (substance A)



170. Select the incorrect statement about NADH during cellular respiration.

- (1) It is synthesized in glycolysis.
- (2) It is transferred into the mitochondria.
- (3) It undergoes oxidative phosphorylation.
- (4) It is reduced to NAD^+ .



171. In Kreb's cycle, GTP is synthesised during ____.

- (1) Conversion of succinyl-CoA to acetyl Co-A
- (2) Conversion of succinyl-CoA to succinic acid
- (3) Conversion of succinyl-CoA to α -ketoglutarate
- (4) Conversion of succinic acid to succinyl- CoA



172. Match the columns I & II and choose the correct option.

Column I		Column II	
A	Aerobic respiration	P	Muscle cells during exercise
B	Alcoholic fermentation	Q	Glycolysis
C	Lactic acid fermentation	R	Yeast
D	Common pathway for aerobic and anaerobic organisms	S	Kreb's cycle

- (1) A-P, B-Q, C-S, D-R
- (2) A-Q, B-P, C-S, D-R
- (3) A-S, B-R, C-Q, D-P
- (4) A-S, B-R, C-P, D-Q



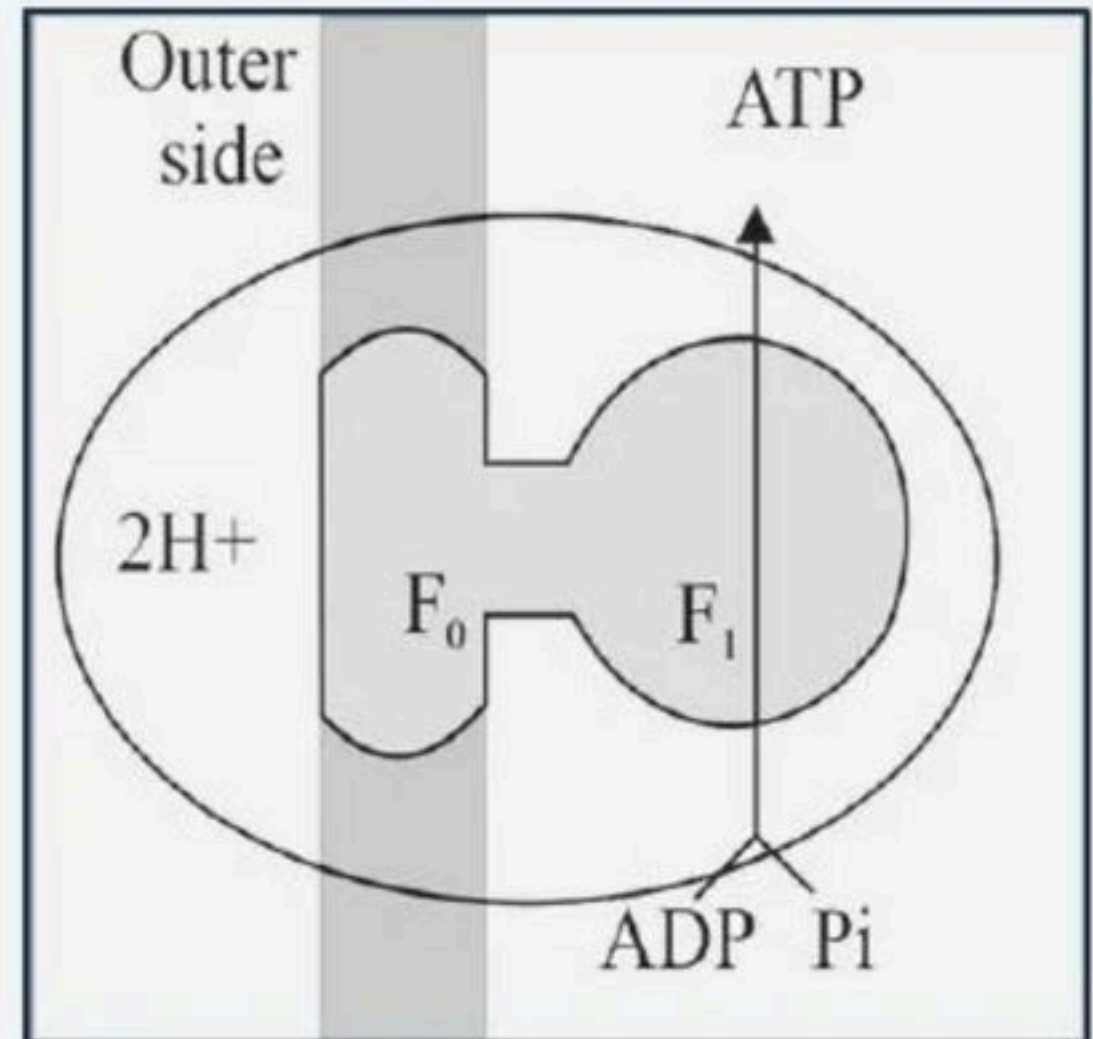
173. Which of the following processes takes place in mitochondria?

- (1) TCA cycle
- (2) ETS
- (3) Carboxylation
- (4) 1 and 2



174. The circled part in the given diagram is part of

- (1) Complex I
- (2) Complex II
- (3) Complex IV
- (4) Complex V



175. Which type of phosphorylation occurs in mitochondria?

- (1) Oxidative phosphorylation
- (2) Photophosphorylation
- (3) 1 and 2
- (4) None of them



176. Ubiquinone transfer its electron to

- (1) Complex III
- (2) Complex II
- (3) Cytochrome c
- (4) Oxygen



177. Match the columns I & II and choose the correct option.

Column I		Column II	
A	In glycolysis, one molecule of glucose yields	P	Acetyl CO-A
B	Oxidative decarboxylation of pyruvic acid yields	Q	Three molecules of CO ₂
C	Complete oxidation of pyruvate in aerobic respiration leaves	R	Two molecules of pyruvate

(1) A-Q, B-R, C-P

(2) A-Q, B-P, C-R

(3) A-R, B-P, C-Q

(4) A-R, B-Q, C-P



178. Identify the incorrect match Compounds on oxidation gives

- (1) Ketoglutaric acid- Succinyl Co-A
- (2) Isocitric acid-Oxalosuccinic acid
- (3) Pyruvic acid - Acetyl Co-A
- (4) Malic acid -Fumaric acid



179. Statement A : In glycolysis, conversion of BPGA to PGA is an energy yielding process, this energy is trapped by the formation of ATP

Statement B : Another ATP is synthesised during the conversion of PEP to pyruvic acid in the glycolytic pathway

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



180. In eukaryotes, most of the Kreb's cycle enzymes are present in

- (1) Chloroplast
- (2) Intermembrane space of mitochondria
- (3) Mitochondrial matrix
- (4) Inner membrane of mitochondria



181. Usable energy available from respiration is approximately

- (1) 55%
- (2) 100%
- (3) 45%
- (4) 50%



182. ____ : Chloroplast ; ____ : Glycolysis ____ : Aerobic respiration

- (1) Photosynthesis, cytoplasm, mitochondria
- (2) Mitochondria, cytoplasm, mitochondria
- (3) Mitochondria chloroplast, cytoplasm
- (4) Mitochondria, cytoplasm, chloroplast



183. Which of the following is correct about cytochrome?

- (1) Small protein attaches to the outer surface of inner membrane of mitochondria.
- (2) Act as non mobile carrier.
- (3) Transfers electron between complex II and III.
- (4) All the above



184. Cellular respiration includes the various pathways by which carbohydrates and other metabolites are broken down with the consecutive buildup of

- (1) ATP
- (2) protein
- (3) vitamins
- (4) fats



185. Statement A : In fermentation there is a net gain of only two molecules of ATP for each molecule of glucose degraded to pyruvic acid

Statement B : Fermentation accounts for complete breakdown of glucose

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



186. Acceptor of acetyl Co-A in Kreb's cycle is

- (1) malic acid
- (2) fumaric acid
- (3) α -keto glutaric acid
- (4) oxaloacetic acid



187. During respiration, yeast converts glucose to

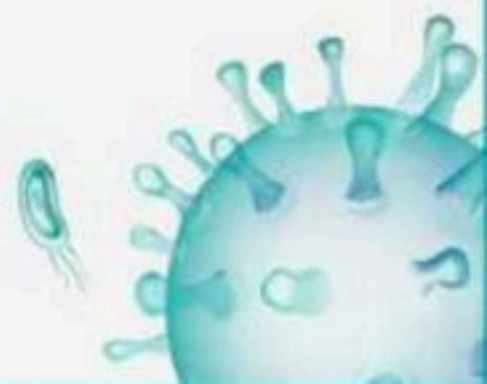
- (1) Ethanol and water
- (2) Ethanol and oxygen
- (3) Ethanol and CO_2
- (4) Lactic acid and CO_2



188. Statement A : All living organisms retain the enzymatic machinery to partially oxidise glucose with the help of oxygen

Statement B : The breakdown of glucose to lactic acid is called glycolysis

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both Statements are correct.
- (4) Both Statements are incorrect.



189. In Kreb's cycle , second CO_2 is liberated during

- (1) Decarboxylation of acetyl Co-A
- (2) Decarboxylation of oxalosuccinate
- (3) Decarboxylation of α -ketoglutarate
- (4) Oxidation of pyruvic acid



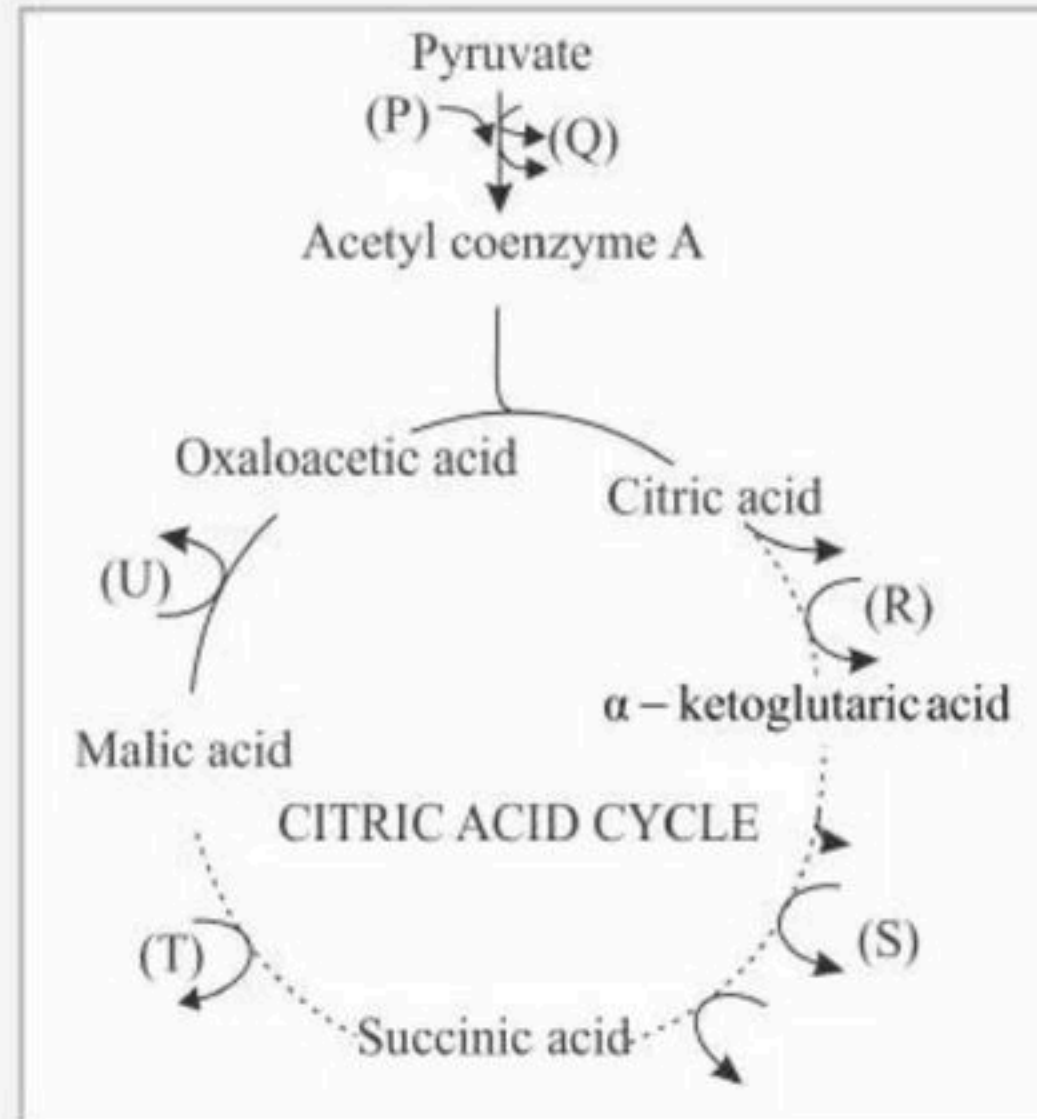
190. What is the role of NAD⁺ in cellular respiration?

- (1) It is a nucleotide source for ATP synthesis.
- (2) It functions as an electron carrier.
- (3) It functions as an enzyme.
- (4) It is the final electron acceptor for anaerobic respiration.



191. In the given diagram label 'U' represents conversion of

- (1) $\text{NADH} + \text{H}^+ \rightarrow \text{NAD}^+$
- (2) $\text{NAD}^+ \rightarrow \text{NADH} + \text{H}^+$
- (3) $\text{FADH}_2 \rightarrow \text{FAD}^+$
- (4) ATP



192. The cytochrome

- (1) Are reduced as they pass electrons on to next molecule
- (2) Located in the matrix
- (3) Has ATP synthase to make ATP
- (4) Are oxidised as they pass electrons on to next molecule



193. Which formula for cellular respiration is correct?

- (1) $1 \text{ glucose} + 6\text{CO}_2 = 6\text{CO}_2 + 6\text{H}_2\text{O} + 38 \text{ ATP}$
- (2) $1 \text{ glucose} + 1\text{O}_2 = 1\text{CO}_2 + 1\text{H}_2\text{O} + 38 \text{ ATP}$
- (3) $1 \text{ glucose} + 6\text{O}_2 = 6\text{CO}_2 + 6\text{H}_2\text{O} + 38 \text{ ATP}$
- (4) $1 \text{ glucose} = 6\text{CO}_2 + 6\text{H}_2\text{O} + 38 \text{ ATP}$



194. Cytochrome is a small protein attached to the

- (1) Inner surface of the outer membrane
- (2) Outer surface of the outer membrane
- (3) Outer surface of the inner membrane
- (4) Inner surface of the inner membrane



195. Match the columns I & II and choose the correct option.

Column I		Column II	
A	Pyruvic acid $\rightarrow$ Acetyl CO-A	P	Formation of GTP molecule
B	Acetyl CO-A $\rightarrow$ Citric acid	Q	Citrate synthase
C	Citric acid $\rightarrow$ Alpha ketoglutaric acid	R	Decarboxylase
D	Alpha ketoglutaric acid $\rightarrow$ Succinic acid	S	Pyruvate dehydrogenase

(1) A-S, B-Q, C-R, D-P

(2) A-R, B-P, C-Q, D-S

(3) A-R, B-Q, C-P, D-S

(4) A-P, B-Q, C-R, D-S



196. Match Column I with Column II :

Choose the correct answer from the options given below :

Column I		Column II	
A	Oxidative decarboxylation	P	Citrate synthase
B	Glycolysis	Q	Pyruvate dehydrogenase
C	Oxidative phosphorylation	R	Electron transport system
D	Tricarboxylic acid cycle	S	EMP pathway

(1) A – R, B – S, C – Q, D – P

(2) A – Q, B – S, C – P, D – R

(3) A – R, B – P, C – Q, D – S

(4) A – Q, B – S, C – R, D – P



197. Match the columns I & II and choose the correct option.

Column I		Column II	
A	Glucose $\rightarrow$ Glucose-6-phosphate	P	Phosphorylation
B	Fructose-6-phosphate $\rightarrow$ Fructose 1,6-bisphosphate	Q	Hexokinase
C	Glyceraldehyde -3-phosphate $\rightarrow$ 1,3 bisphospho glyceric acid	R	Formation of ATP
D	BPGA $\rightarrow$ 3-Phospho glyceric acid	S	Formation of NADH + H ⁺

(1) A-P, B-R, C-Q, D-S

(2) A-S, B-Q, C-R, D-P

(3) A-P, B-Q, C-R, D-S

(4) A-Q, B-P, C-S, D-R



198. Match the columns I & II and choose the correct option.

Column I

A Complex I

B Complex II

C Complex III

D Complex IV

(1) A-Q, B-P, C-R, D-S

(3) A-Q, B-P, C-S, D-R

Column II

P Succinate dehydrogenase

Q NADH dehydrogenase

R Cytochrome c oxidase

S Cytochrome complex

(2) A-R, B-Q, C-S, D-P

(4) A-P, B-Q, C-R, D-S



199. Match the columns I & II and choose the correct option.

Column I		Column II	
A	Net gain of ATP produced in fermentation	P	1
B	Total number of ATP molecules produced in glycolysis	Q	38
C	Net gain of ATP produced in aerobic respiration	R	4
D	Number of FADH_2 produced from one pyruvic acid in Kreb's cycle	S	2

(1) A-P, B-R, C-Q, D-S

(2) A-Q, B-P, C-R, D-S

(3) A-Q, B-R, C-S, D-P

(4) A-S, B-R, C-Q, D-P



200. Match the columns I & II and choose the correct option.

Column I

A 4C-Compound

B 2C-Compound

C 3C-Compound

D 5C-Compound

(1) A-Q, B-R, C-S, D-P

(3) A-R, B-Q, C-S, D-P

Column II

P Acetyl Co-A

Q Pyruvic acid

R Malic acid

S Alpha ketoglutaric acid

(2) A-Q, B-R, C-P, D-S

(4) A-R, B-P, C-Q, D-S



THANK YOU



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